Chemical Bonding Foundations — Lewis Theory, Octet Limitations & Formal Charge

Atoms combine to achieve an inert gas electronic configuration (ns^2 np^6, octet) by sharing, losing, or gaining electrons. Lewis structures account for valence electrons as bonding pairs and lone pairs.

1. Limitations & Failures of the Octet Rule

Octet Exception Type	Characteristics & Electron Count	Classic Chemical Examples
Incomplete Octet (Hypovalent)	Central atom has fewer than 8 valence electrons (electron-deficient)	$BeCl_2$ (4 e-), BF_3 (6 e-), $AlCl_3$ (6 e-), B_2H_6
Expanded Octet (Hypervalent)	Central atom expands coordination using vacant d-orbitals (> 8 e-)	PCl_5 (10 e-), SF_6 (12 e-), IF_7 (14 e-), H_2SO_4 (12 e-)
Odd-Electron Molecules	Unpaired electron present; paramagnetic in gas phase	NO (11 valence e-), NO_2 (17 valence e-), ClO_2 (19 e-)
Noble Gas Compounds	Heavier noble gases form covalent bonds despite full octet	XeF_2, XeF_4, XeF_6, XeO_3, KrF_2

Table 1.1 — Systematic classification of octet rule violations.

2. Formal Charge Calculation

KEY RESULT

Formal Charge (FC) on an atom in a Lewis formula: **FC = V − L − ½ S** where V = valence electrons in free neutral atom, L = non-bonding lone pair electrons, S = shared bonding electrons. · The most stable Lewis structure is the one where formal charges are closest to zero, and negative formal charges reside on the most electronegative atoms.

3. Resonance & Resonance Energy

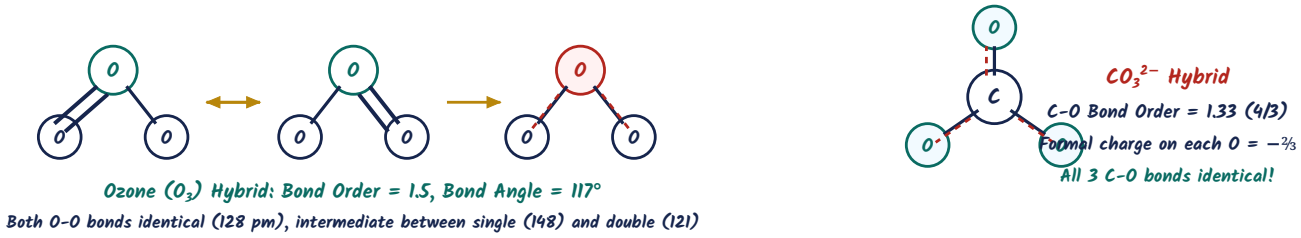


Fig. 1.0 — Lewis octet resonance structures: Ozone O_3 hybrid and Carbonate CO_3^{2-} equivalent C–O bonds.

When a single Lewis structure cannot describe the observed properties of a molecule (e.g. equal bond lengths in CO_3^{2-}, O_3, benzene), the actual molecule is represented as a **resonance hybrid** of canonical contributing structures: · All canonical forms must have identical atomic positions (differ only in electron distribution) and equal unpaired electrons. · **Resonance Energy** = Experimental Heat of Formation − Calculated Heat of most stable canonical form (stabilises the molecule).

The Ionic Bond — Electrostatic Formation, Born-Haber Thermodynamics & Solvation

Formed by complete transfer of electrons between elements of low ionization energy (metals) and high electron affinity (non-metals). Ionic compounds do not exist as discrete molecules, but as infinite crystalline 3D lattices held by nondirectional Coulombic forces.

1. Lattice Energy (U) & Born-Landé Equation

Lattice energy is the energy released when 1 mole of crystalline ionic solid is formed from isolated gaseous ions: **$M^+(g) + X^-(g) \rightarrow MX(s) + U$** Governed by the **Born-Landé Equation**: $U \propto (|z_+ \cdot z_-|) / (r_+ + r_-)$.

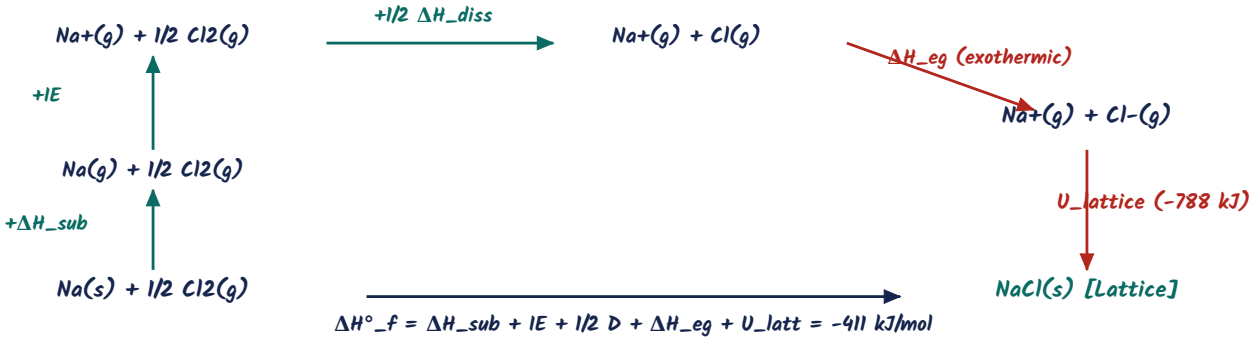


Fig. 1.1 — Born-Haber thermochemical cycle for Sodium Chloride NaCl.

KEY RESULT

Lattice Energy Dependencies: · **Charge Product ($|z_+ \cdot z_-|$):** Dominant factor! E.g. U(MgO) ≈ 4 × U(NaCl) because (+2)(-2) = 4 vs (+1)(-1) = 1. · **Interionic Distance ($r_+ + r_-$):** U(LiF) > U(LiCl) > U(LiBr) > U(LiI).

2. Enthalpy of Solution & Hydration Energy

When an ionic solid dissolves in water: **Δ H_soln = Lattice Energy (U) + Δ H_hyd** · For dissolution to be spontaneous (Δ G U **(exothermic), or entropy of dissolution (TΔ S) overcomes small positive Δ H.** · **If Lattice Energy greatly exceeds Hydration Energy (e.g. BaSO4, CaF2), the compound is** insoluble in water**!**

Fajans' Rules — Polarization, Polarizability & Covalent Character

No ionic bond is 100% ionic. When a small cation approaches a large electron-rich anion, its positive field attracts the anion's electron cloud and repels its nucleus, distorting the electron cloud towards the inter-nuclear region.

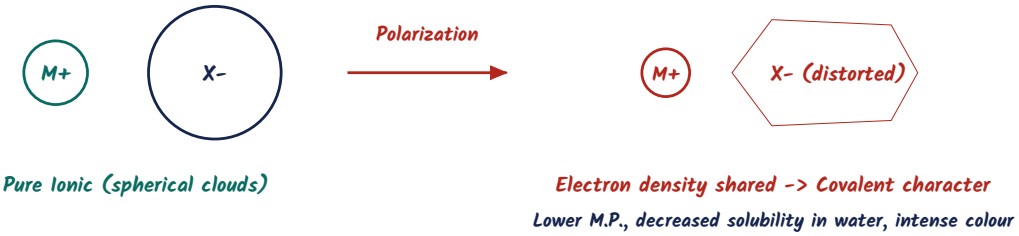


Fig. 1.2 — Polarization of an anion cloud by a polarizing cation leading to covalent character.

1. Fajans' Four Guiding Principles

Factor	Condition Favoring Polarization (Covalency)	Underlying Physical Rationale
1. Cation Size	**Small cation** (High charge density φ = z/r)	Small cation concentrates positive field, exerting intense electrostatic pull on anion cloud (e.g. LiCl is more covalent than KCl)
2. Anion Size	**Large anion** (High polarizability)	Outer valence electrons in large anion are loosely held by nucleus and easily deformed (e.g. AgI is far more covalent than AgF)
3. Ionic Charge	**High charge on Cation or Anion**	Higher positive charge increases pull; higher negative charge makes anion more easily polarizable: $SnCl_4$ (covalent liquid) vs $SnCl_2$ (ionic solid)
4. Electronic Config.	**Pseudo-Noble Gas Cation ($ns^2\ np^6\ ^{nd}10$)**	Cations with 18-electron shell (Cu^+, Ag^+, Zn_2^+, Cd_2^+) have poor d-electron shielding → higher effective Z_{eff} than noble gas cations (Na^+, K^+) of similar size! AgCl is covalent/insoluble, NaCl is ionic/soluble!

Table 1.2 — The four Fajans' rules governing covalent character.

2. Physical Consequences of Polarization in JEE

- **Melting Point Order:** Increased covalency lowers melting points: **NaCl (801°C) > MgCl2 (714°C) > AlCl3 (sublimes 180°C)**; **CaF2 > CaCl2 > CaBr2 > CaI2**.
- **Solubility in Water:** Covalency reduces solubility in polar water and increases solubility in organic solvents (LiCl soluble in alcohol/acetone, NaCl insoluble).
- **Intensity of Colour:** Extensive polarization shifts charge-transfer absorption into the visible spectrum: **AgF (white) → AgCl (white) → AgBr (pale yellow) → AgI (deep yellow)**; **PbO (yellow) vs PbS (black)**.

Solubility & Thermal Stability — Polyatomic Anions & Decomposition Patterns

The thermal stability and aqueous solubility of ionic compounds containing polyatomic oxyanions (CO_3^{2-}, SO_4^{2-}, NO_3^-, OH^-) follow clear thermodynamic principles governed by cation size and lattice energy.

1. Thermal Stability of Carbonates & Nitrates

Thermal decomposition of oxy-salts is driven by the difference in lattice energy between the salt and the product oxide: **MCO_3(s) →{Δ} MO(s) + CO_2(g)** . A small, highly polarizing cation polarizes an adjacent oxygen atom in CO_3^{2-}, weakening the C-O bond and facilitating decomposition into MO + CO2 at lower temperatures!

Oxy-Salt Family	Thermal Stability Order Down Group 1 / 2	Decomposition Reaction & Products
Group 2 Carbonates	**$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$**	$MCO_3 \xrightarrow{(\Delta)} MO + CO_2$ (BaCO3 requires red heat >1300°C)
Group 1 Carbonates	**Li_2CO_3 decomposes; others (Na to Cs) DO NOT decompose!**	$Li_2CO_3 \xrightarrow{(\Delta)} Li_2O + CO_2$ (Li^+ is small like Mg_2+, diagonal relationship). Na_2CO_3 melts without decomposing!
Group 2 Nitrates	**$Be(NO_3)_2 < Mg(NO_3)_2 < Ca(NO_3)_2 < Ba(NO_3)_2$**	$2\ M(NO_3)_2 \xrightarrow{(\Delta)}$ **$2\ MO + 4\ NO_2\uparrow$ (brown) + $O_2\uparrow$**
Group 1 Nitrates	**Decompose to Nitrites (except Li!)**	$2\ NaNO_3 \xrightarrow{(\Delta)}$ **$2\ NaNO_2 + O_2\uparrow$** (No NO2!)n4 $LiNO_3 \xrightarrow{(\Delta)}$ **$2\ Li_2O + 4\ NO_2\uparrow + O_2\uparrow$** (Behaves like Mg!)

Table 1.3 — Thermal stability and decomposition pathways of oxyanion salts.

2. Solubility Trends: Small vs Bulky Anions

- **Rule 1 (Bulky Polyatomic Anions: SO_4^{2-}, CO_3^{2-}, CrO_4^{2-}):** Lattice energy changes very little down the group because r_- is huge ($r_+ + r_- \approx r_-$). Therefore, **hydration energy dominates**! Since hydration energy decreases rapidly with cation size, **solubility DECREASES down the group**: **$BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$ (insoluble!)**.
- **Rule 2 (Small Monatomic Anions: OH^-, F^-):** Lattice energy drops more steeply than hydration energy as cation radius increases. Therefore, **solubility INCREASES down the group**: **$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$**.

Valence Bond Theory (VBT) — Orbital Overlap, Directional Bonding & Bond Strengths

Developed by Heitler and London, extended by Pauling and Slater. A covalent bond is formed by the pairing of spins of electrons in two overlapping half-filled atomic orbitals. Greater overlap = stronger bond!

1. Sigma (σ) vs Pi (π) Bonds

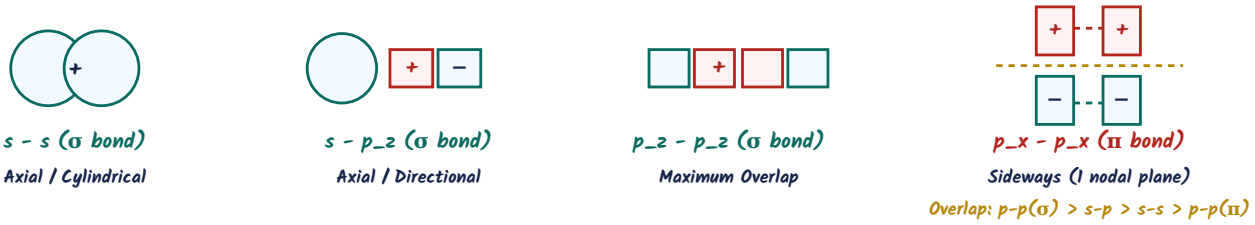


Fig. 1.3 — Orbital overlap types: Axial s-s, s-p, p-p (σ) vs lateral p-p (π).

Feature	Sigma (σ) Bond	Pi (π) Bond
Mode of Overlap	**End-to-end / Axial / Head-on** along inter-nuclear axis	**Sideways / Lateral / Parallel** overlap perpendicular to axis
Types of Overlapping Orbitals	s-s, s-p_z, p_z-p_z, hybrid-hybrid (sp^n-sp^n)	p_x-p_x, p_y-p_y, d_xy-p_y, d-d lateral
Extent of Overlap & Strength	**Greater overlap → Strong bond**	**Lesser overlap → Weaker bond**
Electron Cloud Geometry	Cylindrically symmetrical about inter-nuclear axis	Two sausage-like lobes above and below nodal plane
Rotation Around Bond	**Free rotation** permitted without breaking overlap	**Restricted rotation** (twisting destroys overlap → stereoisomerism!)
Independent Existence	Can exist independently (single bonds are always σ)	Never exists alone; formed only after a σ-bond is established

Table 1.4 — Comprehensive structural contrast between σ and π bonds.

2. Strength Order of Orbital Overlaps

KEY RESULT

For same principal quantum number n: · Axial: **p-p (axial) > s-p > s-s** (p-orbitals are directional, concentrating electron density along bond axis; s-orbitals are spherical and diffuse). · With differing shell n: **1s-1s > 1s-2p > 2s-2p > 2p-2p > 3p-3p** (smaller orbitals overlap closer to nuclei, releasing higher bond energy!).

3. Maximum Overlap Principle & Bond Angles

Pure p-orbitals are oriented at 90° to each other (p_x, p_y, p_z). If bonding occurred via pure p-orbitals without hybridization, bond angles in CH4, NH3, H_2O would all be 90°. The observed angles (109.5°, 107.8°, 104.5°) necessitate the concept of **hybridization**.

Hybridization Theory — Steric Number, Geometry & Percentage s-Character

Hybridization is the mathematical mixing of atomic orbitals of comparable energy on the same atom to form an equal number of new equivalent **hybrid orbitals** having definite directional orientations in space.

1. Steric Number Rule for Hybridization

KEY RESULT

Steric Number (SN) = Number of σ-bonds + Number of lone pairs on central atom · SN = 2 → **sp** (Linear, 180°) · SN = 3 → **sp^2** (Trigonal planar, 120°) · SN = 4 → **sp^3** (Tetrahedral, 109°28′) · SN = 5 → **sp^3d** (Trigonal bipyramidal, axial + equatorial) · SN = 6 → **sp^3d^2** (Octahedral, 90°) · SN = 7 → **sp^3d^3** (Pentagonal bipyramidal, 72° and 90°)

2. Specific d-Orbitals Involved in Hybridization (Critical JEE Advanced Topic)

Hybridization	Ideal Geometry	Specific d-Orbitals Used	Why this d-orbital?
sp^3d	Trigonal bipyramidal (TBP)	**d_{z^2}** (axial bonds lie on z-axis)	d_{z^2} has maximum electron density along z-axis, forming 2 linear axial bonds
dsp^2	Square planar	**$d_{x^2-y^2}$**	Lobes point directly along x and y Cartesian axes to meet ligands
sp^3d^2 / d^2sp^3	Octahedral	**$d_{x^2-y^2}$ and d_{z^2}**	Both e_g orbitals point along Cartesian axes where 6 ligands approach
sp^3d^3	Pentagonal bipyramidal	**d_{z^2}, $d_{x^2-y^2}$, d_{xy}**	d_{z^2} forms 2 axial bonds; $d_{x^2-y^2}$ and d_{xy} lie in xy-equatorial plane

Table 1.5 — Exact d-orbital participation in higher hybrid states.

3. Influence of Percentage s-Character

As percentage s-character increases: sp^3 (25%) sp^2 > sp^3 (s-electrons are held closer to nucleus). · **Bond Angle:** sp (180°) > sp^2 (120°) > sp^3 (109.5°). · **Bond Length:** sp-sp sp^2–sp^2 > sp^3–sp^3.

Bent's Rule — Electronegativity & Hybrid Orbital s/p Character Partitioning

In molecules with non-equivalent hybrid orbitals (e.g. sp^3d or unsymmetrically substituted sp^3), the s- and p-character are not distributed equally among all bonds.

KEY RESULT

Bent's Rule Formulation: "More electronegative substituents prefer hybrid orbitals having **LESS s-character** (more p-character), whereas more electropositive substituents and lone pairs prefer hybrid orbitals having **MORE s-character**."

1. Application to Trigonal Bipyramidal (sp3d) Systems

sp^3d hybridization is partitioned into two independent sets: · **Equatorial Set:** sp^2 hybridized (33.3% s-character, planar 120°). · **Axial Set:** pd_{z^2} hybridized (**0% s-character**, linear 180° along z-axis).

Substituent / Group	Preferred Position in sp^3d	Direct Chemical Consequence in PCl3F2 / SF4
Lone Pairs	**Equatorial** (maximum s-character)	In SF4, lone pair occupies equatorial position (See-Saw shape). In ClF3, 2 lone pairs are equatorial (T-shaped). In XeF2, 3 lone pairs are equatorial (Linear).
Electronegative Atoms (F)	**Axial** (minimum s-character)	In PCl3F2, both Fluorine atoms go strictly to **axial** positions! Molecule is non-polar (μ = 0).
Less Electronegative / Bulky (CH3, Cl)	**Equatorial** (higher s-character)	In $P(CH_3)_2F_3$, both CH3 groups occupy equatorial sites.

Table 1.6 — Applications of Bent's Rule in sp3d stereochemistry.

2. Bond Angle Modifications in sp3 Molecules

- **CH2F2 vs CH4:** In CH2F2, the electronegative F atoms pull p-character into the C-F bonds. Consequently, the C-H bonds gain extra s-character, causing the **H-C-H bond angle to EXPAND (>109.5°)** and the **F-C-F bond angle to CONTRACT (<109.5°)**!
- **Bond Length Effect:** Higher s-character shortens the bond: C-H bond in CH2F2 is shorter and stronger than in CH4.

Drago's Rule — Ineffective Hybridization in Heavier Period 3+ Hydrides

Why are the bond angles in PH_3, AsH_3, H_2S, H_2Se all close to 90°, while NH_3 is 107.8° and H_2O is 104.5°? Drago's rule provides the thermodynamic explanation.

KEY RESULT

Drago's Rule Conditions: Hybridization does **NOT take place** if all three conditions are satisfied: 1. The central atom belongs to **Period 3 or below** (P, As, Sb, Bi, S, Se, Te). 2. The central atom has at least **one non-bonding lone pair**. 3. The attached terminal atoms have electronegativity $\chi \leq$ **2.1** (e.g. Hydrogen, χ_H = 2.1).

1. Bond Angle Comparison: Period 2 vs Periods 3-5

Group 15 Hydride	Bond Angle	Group 16 Hydride	Bond Angle	Stereochemical Status
NH_3	**107.8°**	H_2O	**104.5°**	Normal sp^3 hybridization with lone pair repulsions
PH_3	**93.5°**	H_2S	**92.1°**	Drago system: No hybridization, pure p-orbital bonds
AsH_3	**91.8°**	H_2Se	**91.0°**	Pure p-orbital overlap (angle approaches 90°)
SbH_3	**91.3°**	H_2Te	**89.5°**	Pure p-orbital overlap (angle approaches 90°)

Table 1.7 — Sharp bond angle drop illustrating Drago's rule.

2. Chemical Consequences of Drago's Rule

- **Lone Pair Nature:** In PH_3 and H_2S, the lone pair resides in an **almost pure unhybridized s-orbital** (spherical, close to nucleus, high s-character).
- **Lewis Basicity:** NH_3 is a strong Lewis base because its lone pair is in an sp^3 directional orbital pointing outwards. PH_3 is an **extremely poor Lewis base** because its s-orbital lone pair is tightly held and nondirectional!
- **Proton Affinity:** PH_3 does not protonate easily; PH_4^+ exists only with conjugate bases of superacids (PH_4I), whereas NH_4^+ salts are ubiquitous.

VSEPR Theory — Gillespie-Nyholm Electron Pair Repulsions & Molecular Geometries

Valence Shell Electron Pair Repulsion (VSEPR) predicts molecular geometry by minimizing electrostatic repulsions between electron pairs (bonding pairs and lone pairs) surrounding the central atom.

KEY RESULT

Repulsion Hierarchy: lp − lp > lp − bp > bp − bp · Lone pairs occupy more space because they are attracted by only one positive nucleus, whereas bond pairs are shared between two nuclei.

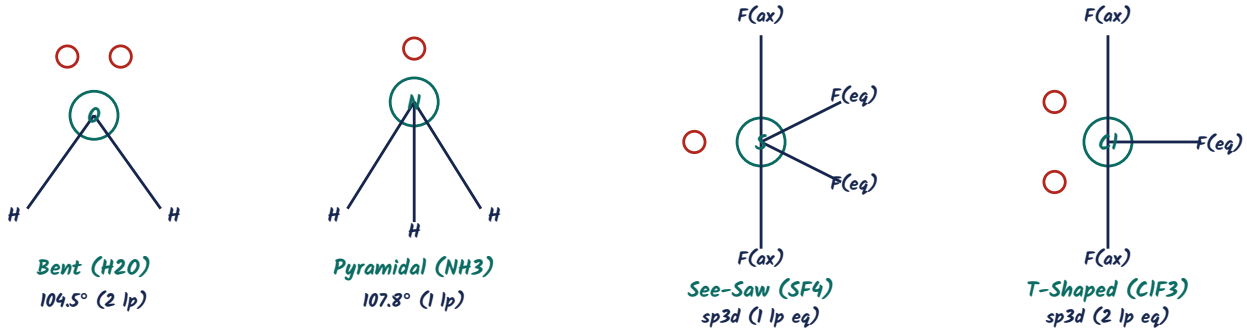


Fig. 2.1 — Key distorted VSEPR geometries: Bent (H_2O), Pyramidal (NH3), See-Saw (SF4), T-Shaped (ClF3).

1. Comprehensive VSEPR Geometries Table

Type	Formula	Bond Pairs	Lone Pairs	Electron Geometry	Observed Molecular Shape	Examples
AB2	BeCl2	2	0	Linear	**Linear** (180°)	BeF2, CO2, HCN, CS2
AB2E	SO2	2	1	Trigonal planar	**Bent / V-shaped** (<120°)	SO2, O_3, NO_2^-, SnCl2(g)
AB3	BF3	3	0	Trigonal planar	**Trigonal planar** (120°)	BF3, BCl3, SO3, NO_3^-
AB4	CH4	4	0	Tetrahedral	**Tetrahedral** (109.5°)	CH4, CCl4, SiF4, NH_4^+, SO_4^{2-}
AB3E	NH3	3	1	Tetrahedral	**Trigonal pyramidal** (107.8°)	NH3, PCl3, H_3O^+, ClO_3^-
AB2E2	H_2O	2	2	Tetrahedral	**Bent / V-shaped** (104.5°)	H_2O, OF2, SCl2, ClO_2^-
AB5	PCl5	5	0	Trigonal bipyramidal	**Trigonal bipyramidal**	PCl5, PF5, AsF5
AB4E	SF4	4	1	Trigonal bipyramidal	**See-Saw** (1 lp eq)	SF4, SeF4, TeCl4
AB3E2	ClF3	3	2	Trigonal bipyramidal	**T-Shaped** (2 lp eq)	ClF3, BrF3, ICl3
AB2E3	XeF2	2	3	Trigonal bipyramidal	**Linear** (3 lp eq, 180°)	XeF2, I_3^-, ICl_2^-
AB6	SF6	6	0	Octahedral	**Regular Octahedral** (90°)	SF6, SeF6, $[PF_6]^-$
AB5E	BrF5	5	1	Octahedral	**Square Pyramidal**	BrF5, IF5, XeOF4
AB4E2	XeF4	4	2	Octahedral	**Square Planar** (2 lp ax)	XeF4, $[ICl_4]^-$
AB7	IF7	7	0	Pentagonal bipyramidal	**Pentagonal bipyramidal**	IF7 (72° eq, 90° ax)
AB6E	XeF6	6	1	Pentagonal bipyramidal	**Distorted Octahedral**	XeF6 (fluxional lp)

Table 2.1 — Master compendium of VSEPR molecular geometries.

Bond Parameters — Energetics, Radii, Resonance & Angle Determinants

A covalent bond is characterized by four interrelated quantitative parameters: Bond Order, Bond Length, Bond Dissociation Enthalpy, and Bond Angle.

1. Bond Order vs Length vs Dissociation Enthalpy

KEY RESULT

General Interrelationships: · **Bond Order** ∝ **Bond Strength** ∝ **Force Constant** ∝ **Bond Energy** · **Bond Order** ∝ **1 / (Bond Length)** · E.g. Carbon-Carbon bonds: C-C (154 pm, 348 kJ) → C=C (134 pm, 614 kJ) → C≡ C (120 pm, 839 kJ).

2. Master Rules Governing Bond Angles in Chemistry

Determinant Factor	Governing Rule / Trend	Classic Comparative Examples
1. Hybridization State	Higher s-character opens the angle: sp (180°) > sp^2 (120°) > sp^3 (109.5°)	BeCl2 (180°) > BCl3 (120°) > CH4 (109.5°)
2. Number of Lone Pairs	Each additional lone pair compresses bond angle by ~2° to 2.5°	CH4 (0 lp, 109.5°) > NH3 (1 lp, 107.8°) > H_2O (2 lp, 104.5°)
3. Central Atom Electronegativity (Same Halogen/Terminal)	When terminal atoms are identical, **higher central atom EN increases bond angle** (electron pairs pulled closer to center → greater mutual repulsion)	NH3 (107.8°) > PH3 (93.5°) > AsH3 (91.8°)nH2O (104.5°) > H_2S (92.1°) > H_2Se (91.0°)
4. Terminal Atom Electronegativity (Same Central Atom)	When central atom is identical, **higher terminal atom EN decreases bond angle** (electron pairs pulled away from center → lesser mutual repulsion)	PI3 (102°) > PBr3 (101.5°) > PCl3 (100°) > PF3 (97.8°)nOF2 (103.2°) < H_2O (104.5°) < Cl_2O (111°!)
5. Steric Crowding of Bulky Groups	Bulky ligands force angles wide open to relieve van der Waals clash	$O(CH_3)_2$ (111.7°) > H_2O (104.5°); $O(SiH_3)_2$ (144°!)

Table 2.2 — Systematic hierarchy determining bond angle variations.

Dipole Moment (μ) — Polar Bonds, Molecular Geometry & % Ionic Character

A bond between two atoms of different electronegativity is polar. The electric dipole moment measures the extent of charge separation: $\{\mu\} = q \times d$ (units: Debye, 1 D = 3.33564 × 10⁻30 C·m).

1. Molecular Dipole Moment as a Vector Sum

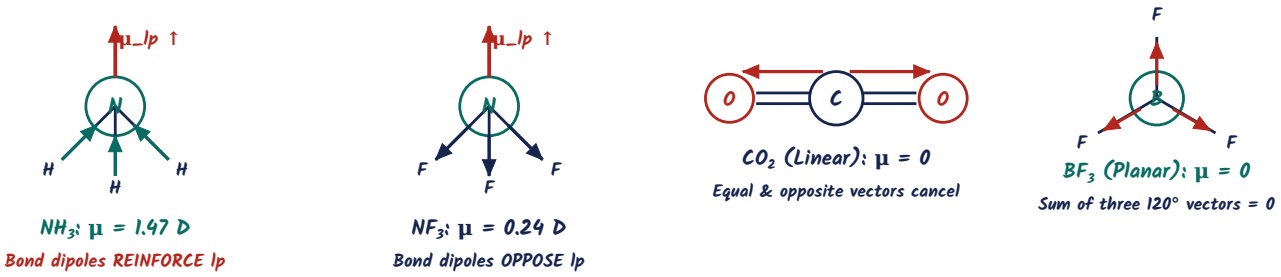


Fig. 2.2 — Dipole moment vector analysis: NH3 vs NF3 and planar symmetrical cancellation.

Molecular dipole moment is the **vector resultant** of all individual bond dipole moments and lone pair moments: $\mu_{net} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2 \cos\theta}$. Symmetrical molecules with identical surrounding atoms have **μ = 0** (vectors cancel): Linear (CO2, BeF2), Trigonal planar (BF3), Tetrahedral (CCl4, CH4), Octahedral (SF6), Square planar (XeF4).

2. The Classic NH3 vs NF3 Comparison (JEE Favorite)

Feature	Ammonia (NH3)	Nitrogen Trifluoride (NF3)
Individual Bond Dipoles	N is more EN than H: $N^{\delta-} \leftarrow H^{\delta+}$ (bond moments point **TOWARDS** N)	F is more EN than N: $N^{\delta+} \rightarrow F^{\delta-}$ (bond moments point **AWAY** from N)
Lone Pair Dipole Direction	Lone pair points upwards away from N nucleus	Lone pair points upwards away from N nucleus
Vector Resultant	**Additive:** Orbital dipole and 3 N-H bond dipoles reinforce in the same direction!	**Subtractive:** Lone pair dipole opposes the resultant of the 3 N-F bond dipoles!
Experimental Net Dipole (μ)	**1.47 D (High)**	**0.24 D (Very Low!)**

Table 2.3 — Why Ammonia has a six-fold higher dipole moment than Nitrogen Trifluoride.

3. Percentage Ionic Character (Hannay-Smith Equation)

KEY RESULT

% Ionic Character = 16 $|\chi_A - \chi_B|$ + 3.5 $(\chi_A - \chi_B)^2$ Alternatively, from dipole measurements: **% Ionic Character = $\{\mu_{obs}\}\{\mu_{calc}\}$ × 100%**

Back-Bonding (Dative π-Bonding) — Lewis Acidity & Structural Reorganisation

Back-bonding is an internal coordinate π-bond formed when a filled p-orbital of a donor atom (with lone pairs) overlaps sideways with a vacant unhybridized p- or d-orbital on an adjacent electron-deficient acceptor atom.

1. Types of Back-Bonding

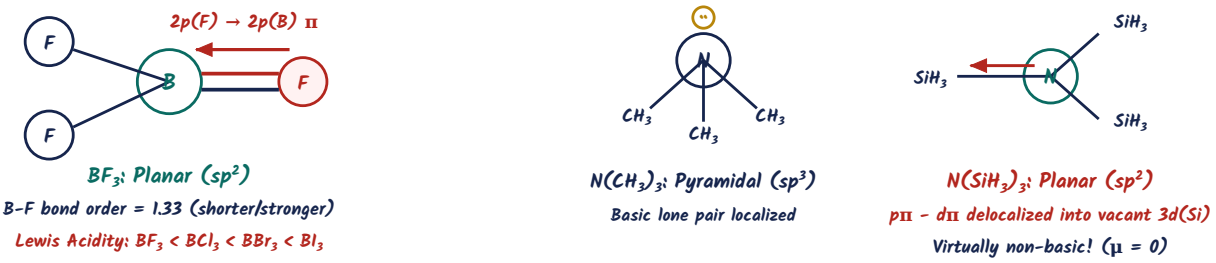


Fig. 2.3 — Dative back-bonding: BF3 pπ-pπ backbond and $N(SiH_3)_3$ planarization via pπ-dπ.

Type of Back-Bonding	Orbital Interaction	Key Chemical Examples	Major Observed Consequences
pπ - pπ Back-Bonding	Filled 2p (donor) → Empty 2p (acceptor)	BF3, $B(OMe)_3$, $B_3N_3H_6$	B-F bond length drops to 130 pm (partial double bond); Inversion of BX3 Lewis acidity: **BF3 < BCl3 < BBr3 < BI3**
pπ - dπ Back-Bonding	Filled 2p (donor) → Empty 3d (acceptor)	$N(SiH_3)_3$, $O(SiH_3)_2$, P_4O_{10}, SO2, SO3	Planarization of amine, drastic decrease in basicity, bond angle widening

Table 2.4 — Classification and manifestations of back-bonding.

2. Trimethylamine N(CH3)3 vs Trisilylamine N(SiH3)3 (Classic JEE Topic)

Property	Trimethylamine $N(CH_3)_3$	Trisilylamine $N(SiH_3)_3$
Geometry around N	**Trigonal Pyramidal** (sp^3 N)	**Planar Trigonal** (sp^2 N)
Back-Bonding	**None** (Carbon has no empty d-orbitals)	**Extensive 2pπ(N) → 3dπ(Si) back-bonding**
Hybridization of N	sp^3 (lone pair resides in hybrid orbital)	sp^2 (lone pair delocalised into 3d orbitals of 3 Si atoms)
Lewis Basicity	**Strong Lewis base** (donates available lone pair)	**Weak / Non-basic** (lone pair is delocalised and unavailable)
Adduct with BMe3	Forms stable adduct $Me_3N{:}BMe_3$	Does not form adduct (sterically and electronically inert)

Table 2.5 — Structural contrast caused by pπ-dπ back-bonding.

KEY RESULT

*Inversion of Lewis Acidity in Boron Trihalides: **BF3 < BCl3 < BBr3 < BI3**. Reason: 2p(B) <--- 2p(F) overlap is optimal (same size and energy), satisfying B's electron deficiency. In BI3, 2p(B) <--- 5p(I) overlap is ineffective due to size/energy mismatch, leaving B completely hungry for external electron pairs!*

Bridge Bonding (Multicentre Bonds) — 3c-2e Banana Bonds vs 3c-4e Halide Bridges

When monomeric molecules have incomplete octets (electron-deficient) and cannot satisfy them by back-bonding, they dimerise or polymerise through multicentre bridging bonds.

1. 3c-2e 'Banana' Bonds (Electron-Deficient Bridges)



Fig. 2.4 — Bridge bonding modes: Diborane B_2H_6 3c-2e banana bond vs Al_2Cl_6 3c-4e dative bridge.

- **Diborane (B_2H_6):** Two B atoms and four terminal H atoms are coplanar. Two bridging H atoms lie above and below this plane. Each B-Hb-B bridge is a **3-centre 2-electron (3c-2e)** bond sharing 2 electrons over 3 nuclei (B, H, B). Each Boron atom is **sp^3 hybridized** (12 valence electrons total).
- **Beryllium Hydride (BeH2)_n:** Solid (BeH2)_n forms an infinite chain of Be atoms joined by 3c-2e Be-H-Be bridges (Be is sp^3 hybridized).
- **Aluminium Alkyls [Al(CH_3)$_3$]2:** Dimerises via **3c-2e Al-C-Al bridges** where the methyl carbon uses an sp^3 orbital to bridge two Al atoms.

Feature	3c-2e Bridge (e.g. B_2H_6, [AlMe3]2)	3c-4e Bridge (e.g. Al_2Cl_6, I_2Cl_6)
Bridging Atom	Hydrogen (H) or Methyl (CH3) without lone pairs	Halogen (Cl, Br, I) having lone pairs
Total Electrons in Bridge	**2 electrons** (electron-deficient 'banana' bond)	**4 electrons** (1 normal covalent bond + 1 dative coordinate bond)
Lewis Acidity of Dimer	Still somewhat electron-deficient (cleaved by Lewis bases)	Octet of central metal is completely satisfied
Bond Order of Bridge	0.5 per bridge link (weaker than terminal)	1.0 per coordinate/covalent bond

Table 2.6 — Comparison between 3c-2e and 3c-4e bridging architectures.

2. 3c-4e Halogen Bridges (Al2Cl6, BeCl2)

In **Al_2Cl_6**, each bridging chlorine atom donates a lone pair into the empty 3p/3d orbital of the adjacent Al atom. Each Al atom achieves an octet through two terminal covalent bonds and two bridging (one covalent, one dative) bonds (sp^3 Al). In **BeCl2(solid)**, an infinite linear chain is formed via Be-Cl: → Be coordinate bridges (sp^3 Be, 4-coordinate). In vapor at 1200 K, it exists as linear monomer Cl-Be-Cl (sp Be).

Molecular Orbital Theory (MOT) — LCAO Principles, Bonding/Antibonding & s-p Mixing

Developed by Hund and Mulliken. Atomic orbitals combine linearly (LCAO) to form **Molecular Orbitals (MOs)** delocalised over the entire molecule. Total number of MOs formed equals total number of combining atomic orbitals.

1. Bonding vs Antibonding Molecular Orbitals

Feature	Bonding MO ($\psi_b = \psi_A + \psi_B$)	Antibonding MO ($\psi_a = \psi_A - \psi_B$)
Formation Mode	Constructive interference (in-phase wave addition)	Destructive interference (out-of-phase wave subtraction)
Electron Density	**High between nuclei** (shields nuclei from each other)	**Zero at nodal plane** between nuclei
Energy State	**Lower energy** than combining atomic orbitals (Stabilising)	**Higher energy** than combining atomic orbitals (Destabilising)

Table 2.7 — Bonding versus Antibonding orbital characteristics.

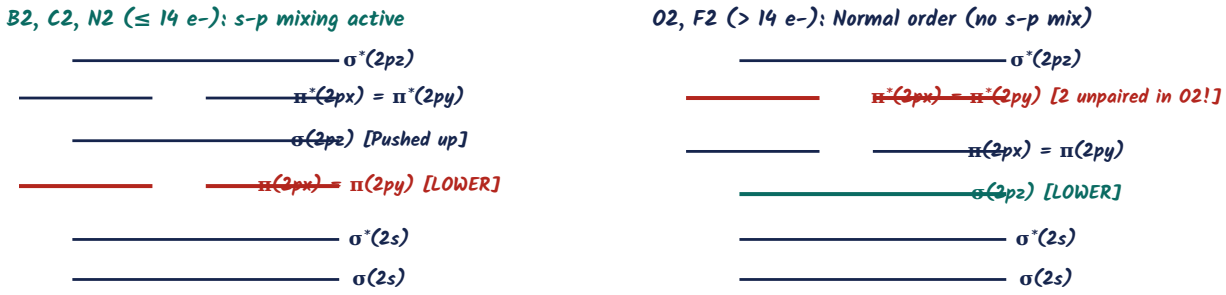


Fig. 2.2 — MO energy level crossover caused by s-p mixing: B_2, C_2, N_2 (≤14 e-) vs O_2, F_2 (>14 e-).

2. The s-p Mixing Phenomenon (The Crossover in Energy)

In homonuclear diatomics of Li, Be, B, C, N (≤ 14 electrons), the energy gap between 2s and 2p_z atomic orbitals is small (14e⁻ (O_2, F_2): Large 2s-2p gap (> 15 eV) prevents mixing; normal order where σ(2p_z) is lower than π(2p). **Bond Order (BO) = $(N_b - N_a) / 2$**

Homonuclear Diatomics — Complete Electronic Configurations, Bond Orders & Magnetism

Evaluation of 2nd-row homonuclear diatomics using MO theory.

1. Master Electronic Configurations Table

Molecule / Ion	Total e-	MO Electronic Configuration	Bond Order	Magnetic Character
H_2	2	σ(1s)^2	**1**	Diamagnetic
He_2+	3	$\sigma(1s)^2\ \sigma^*(1s)^1$	**0.5**	Paramagnetic (1 unpaired e-)
He_2	4	$\sigma(1s)^2\ \sigma^*(1s)^2$	**0**	Does NOT exist!
B_2	10	KK $\sigma(2s)^2\ \sigma^*(2s)^2$ [$\pi(2p_x)^1 = \pi(2p_y)^1$]	**1**	**Paramagnetic** (2 unpaired e- in degenerate π)
C_2	12	KK $\sigma(2s)^2\ \sigma^*(2s)^2$ [$\pi(2p_x)^2 = \pi(2p_y)^2$]	**2**	**Diamagnetic** (Both bonds are **pure π-bonds!**)
N_2	14	KK $\sigma(2s)^2\ \sigma^*(2s)^2$ [$\pi(2p_x)^2 = \pi(2p_y)^2$] $\sigma(2p_z)^2$	**3**	Diamagnetic (1 σ + 2 π bonds)
N_2+	13	KK $\sigma(2s)^2\ \sigma^*(2s)^2$ [$\pi(2p_x)^2 = \pi(2p_y)^2$] $\sigma(2p_z)^1$	**2.5**	Paramagnetic (BO: N_2 > N_2+)
O_2+	15	KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2$ [$\pi(2p)^4$] [$\pi(2p_x)^1$]	**2.5**	Paramagnetic (BO: O_2 + > O_2)
O_2	16	KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2$ [$\pi(2p)^4$] [$\pi(2p_x)^1 = \pi^*(2p_y)^1$]	**2**	**Paramagnetic** (2 unpaired e- in π)
O_2^- (Superoxide)	17	KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2$ [$\pi(2p)^4$] [$\pi(2p)^3$]	**1.5**	Paramagnetic (1 unpaired e-)
O_2^{2-} (Peroxide)	18	KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2$ [$\pi(2p)^4$] [$\pi(2p)^4$]	**1**	Diamagnetic (BO = 1, single O-O bond)
F_2	18	KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2$ [$\pi(2p)^4$] [$\pi(2p)^4$]	**1**	Diamagnetic
Ne_2	20	KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2$ [$\pi(2p)^4$] [$\pi(2p)^4$] $\sigma^*(2p_z)^2$	**0**	Does NOT exist!

Table 2.8 — Complete MO properties of homonuclear diatomics.

KEY RESULT

Two Critical JEE Advanced Insights from MOT: 1. **C_2 molecule:** In gaseous C_2, the double bond consists of **TWO PI (π) BONDS AND NO SIGMA BOND** because the last 4 electrons enter the degenerate π(2p_x) and π(2p_y) orbitals! 2. **Bond Length Order of Dioxygen species:** Bond Order: O_2^+ (2.5) > O_2 (2.0) > O_2^- (1.5) > O_2^{2-} (1.0) Bond Length: **$O_2^+ < O_2 < O_2^- < O_2^{2-}$** (Peroxide has longest bond, ~149 pm).

Heteronuclear Diatomics — Polar Molecular Orbitals, CO Anomaly & NO Redox

In heteronuclear diatomics (AB), the more electronegative atom has lower atomic orbital energies, contributing more to bonding MOs, while the electropositive atom contributes more to antibonding MOs.

1. Carbon Monoxide (CO) — The HOMO/LUMO Anomaly

- **Isoelectronic with N_2:** 14 valence electrons (10 outer). Ground state configuration: $\sigma(2s)^2\ \sigma^*(2s)^2\ [\pi(2p_x)^2 = \pi(2p_y)^2]\ \sigma(2p_z)^2$ Bond order = (8 - 2)/2 = 3 (one σ, two π bonds).
- **The CO+ Ionization Anomaly (Classic JEE Advanced Question):** When CO is ionised to CO+, an electron is removed from the HOMO σ(2s). However, due to extensive s-p mixing and hybridization, this HOMO has **slight antibonding character**! Consequently, upon removing this electron: **Bond Order of CO+ INCREASES to ~3.5**, and the C-O bond becomes **SHORTER AND STRONGER** (112.8 pm → 111.5 pm)!

2. Nitric Oxide (NO) — Odd-Electron Diatomic

Total valence electrons = 15. Electronic configuration: **KK $\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ [\pi(2p_x)^2 = \pi(2p_y)^2]\ [\pi^*(2p_x)^1]$** · Bond Order = (8 - 3)/2 = 2.5. Paramagnetic (1 unpaired electron in π^*). · Oxidation to Nitrosonium ion **NO+**: Removing the antibonding π electron yields NO+ (isoelectronic with N_2): **Bond Order: NO^+ (3.0) > NO (2.5) > NO^- (2.0)** NO+ is diamagnetic and has a shorter, much stronger bond than neutral NO.

Hydrogen Bonding — Dipole-Dipole Attraction, Conditions & Energetics

A special electrostatic attraction between a hydrogen atom covalently bonded to a strongly electronegative small atom (F, O, N) and a lone pair on another electronegative atom: $X\text{-}H^{\delta+} \cdots :Y^{\delta-}$. Bond strength ranges from 8 to 42 kJ/mol (much weaker than covalent 400 kJ, but stronger than van der Waals 2-4 kJ).

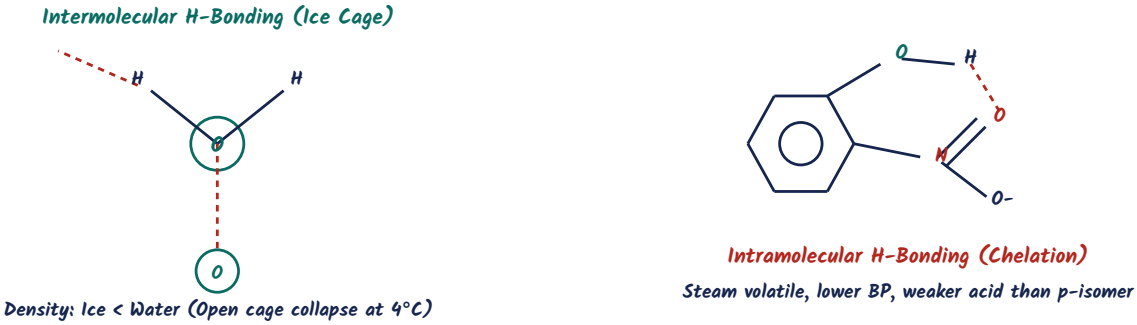


Fig. 3.1 — Intermolecular H-bonding in ice cage vs Intramolecular chelation in o-nitrophenol.

1. Intermolecular vs Intramolecular Hydrogen Bonding

Feature	Intermolecular H-Bonding	Intramolecular H-Bonding (Chelation)
Definition	Occurs between **different molecules** of same or different substances	Occurs **within the same molecule** between two functional groups
Ring Formation	Forms open chains or 3D polymeric networks	Forms a stable **5- or 6-membered chelate ring**
Boiling / Melting Point	**Significantly ELEVATES** M.P. and B.P. (requires energy to break intermolecular network)	**LOWERS** B.P. and M.P. (molecule curls internally, preventing intermolecular association)
Viscosity & Surface Tension	**High viscosity** (e.g. glycerol, H_2SO_4, water)	**Low viscosity**, high volatility (steam volatile)
Classic Organic Example	**p-Nitrophenol** (high BP, soluble in water, non-volatile)	**o-Nitrophenol** (steam volatile, lower BP, less soluble in water)

Table 3.1 — Comparison of Inter- vs Intramolecular Hydrogen Bonding.

2. Physical Anomalies Caused by H-Bonding in Water

- **Liquid Water vs Ice Density (Maximum Density at 4°C):** In ice, each oxygen is tetrahedrally surrounded by four H atoms (two covalent, two H-bonds), creating an **open cage-like structure** with large hexagonal voids. Density of ice < water (ice floats!). On heating from 0°C to 4°C, the cage collapses, packing molecules closer → **Density reaches maximum at 4°C (277 K)**.
- **Boiling Points of Group 15, 16, 17 Hydrides:** H_2O, HF, NH3 have abnormally high boiling points compared to their heavier group congeners (H_2S, HCl, PH3) due to strong intermolecular H-bonding.

Symmetrical Hydrogen Bonds & Clathrates — Strongest H-Bonds & Ice Clathrates

While most hydrogen bonds are unsymmetrical (X-H ··· Y, length ~170 pm), exceptionally strong **symmetrical hydrogen bonds** exist where the proton sits exactly equidistant between two electronegative atoms.

1. The Bifluoride Ion [HF2]- & KHF2 Salt

- **Bifluoride Ion [F-H-F]⁻:** When KF dissolves in anhydrous liquid HF, it forms **Potassium bifluoride KHF2** containing the linear [F-H-F]⁻ anion: **KF + HF → K⁺ [F-H-F]⁻** · The proton sits symmetrically in the center: both F-H distances are identical **(113 pm)**. · Bond energy = **~165 kJ/mol** (the strongest hydrogen bond known, possessing significant covalent character!).
- **Non-Existence of KHCl2, KHBr2, KHI2:** Chlorine, Bromine, and Iodine cannot form [HCl2]⁻ because their lower electronegativity and larger radii make H-bonding far too weak to stabilize the triatomic complex anion.

2. Clathrate Hydrates (Gas Hydrates)

Type of Clathrate	Host Framework	Guest Trapped Atoms	Key Conditions & Properties
Methane Clathrate (Burning Ice)	Ice hydrogen-bonded cage (tetrakaidecahedral)	Methane CH4 gas molecules	Forms on ocean floors under high pressure; massive fossil energy reserve: $CH_4 \cdot 5.75\ H_2O$
Noble Gas Clathrates	Quinol (hydroquinone) hydrogen-bonded cage	Ar, Kr, Xe (atomic radius 1.9 – 2.2 Å)	He and Ne are too small and escape through the cage pores; no chemical bonding involved!

Table 3.2 — Hydrogen-bonded clathrate and inclusion architectures.

KEY RESULT

TRAP: Does KHF2 exist as $K^+ + H^+ + 2\ F^-$? NO! It exists strictly as K^+ and the complex bifluoride anion $[HF_2]^-$. On heating, it decomposes into KF + HF↑.

van der Waals Forces — Classification of Weak Non-Covalent Interactions

Collective term for weak intermolecular attractive forces that operate between neutral molecules. They are non-directional and decay rapidly with distance (typically as 1/r^6).

1. Three Components of van der Waals Forces

Interaction Name	Nature of Interacting Molecules	Distance Dependence	Physical Mechanism
1. Keesom Forces (Dipole-Dipole)	**Two permanent polar dipoles** (e.g. HCl···HCl)	$E \propto 1/r^3$ (stationary), $E \propto 1/r^6$ (rotating)	Electrostatic alignment of permanent positive and negative ends of polar molecules (temperature dependent)
2. Debye Forces (Dipole-Induced Dipole)	**Permanent polar dipole + Non-polar molecule** (e.g. HCl···noble gas)	$E \propto 1/r^6$	Permanent dipole induces an instantaneous dipole in the polarizable electron cloud of adjacent non-polar molecule
3. London Dispersion Forces (Instantaneous Induced Dipole)	**Two non-polar molecules / Noble gases** (e.g. He···He, CH_4···CH_4, I_2 ···I_2)	$E \propto 1/r^6$	Synchronized quantum fluctuations of electron density create transient dipoles that induce corresponding dipoles in neighbors

Table 3.3 — The three van der Waals interaction regimes.

2. Factors Influencing London Dispersion Forces

- **Molecular Size / Surface Area:** Larger molecules have greater surface area of contact → stronger dispersion forces (e.g. Boiling points: n-pentane (36°C) > isopentane (28°C) > neopentane (9.5°C) due to spherical branching).
- **Polarizability (Number of Electrons):** Heavier atoms have diffuse, easily deformable electron clouds → stronger forces: He < Ne < Ar < Kr < Xe and F_2 < Cl_2 < Br_2 < I_2 (gas → liquid → solid!).

Metallic Bonding — Drude-Lorentz Electron Sea & Band Theory of Solids

Metals possess low ionization energies and vacant valence orbitals. Metallic bonding accounts for their high electrical and thermal conductivity, metallic luster, malleability, and ductility.

1. Electron Sea (Drude-Lorentz) Model

A metal is viewed as a regular lattice of positive metal ions (kernels) immersed in a delocalised 'sea' of mobile valence electrons. Nondirectional electrostatic attraction binds kernels to the electron sea: · **Electrical Conductivity:** Applied electric field causes directional drift of mobile electrons. · **Malleability & Ductility:** When hammered, layers of metal kernels slide over one another without shattering because the electron sea instantly adjusts to new positions (unlike brittle ionic crystals).

2. Band Theory of Solids (Molecular Orbital Continuum)

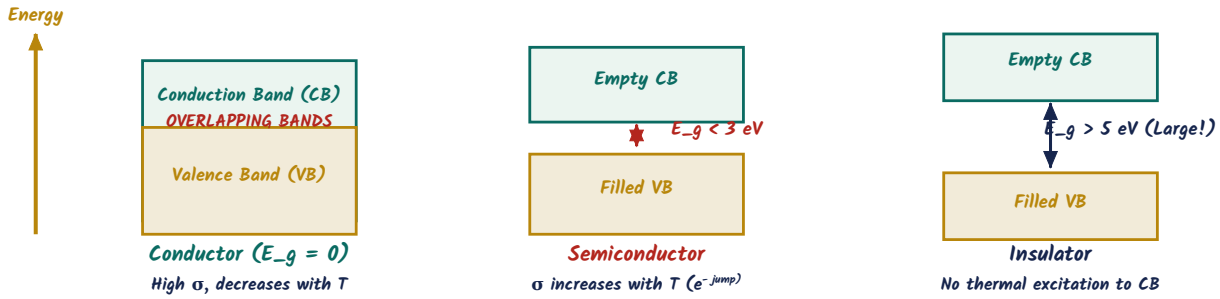


Fig. 3.2 — Band theory: Conductor (overlapping bands), Semiconductor (small gap), Insulator (large gap).

In an infinite crystal lattice of N atoms, N atomic orbitals overlap to form an almost continuous band of energy levels: · **Valence Band (VB):** Lower energy band filled with valence electrons. · **Conduction Band (CB):** Higher energy unoccupied band. · **Forbidden Energy Gap (E_g):** Energy separation between top of VB and bottom of CB.

Solid Type	Energy Bandgap (E_g)	Conduction Mechanism & Temperature Dependence
Conductor (Metal)	**E_g = 0** (Valence band overlaps conduction band)	Electrons move freely into conduction band. **Conductivity DECREASES with increasing T** (lattice phonon vibrations scatter electrons)
Semiconductor (Si, Ge)	**Small E_g (0.5 to 3 eV)**	Thermal excitation promotes electrons across gap. **Conductivity INCREASES with increasing T** (intrinsic and extrinsic doping: p-type and n-type)
Insulator (Diamond, Quartz)	**Large E_g (> 5 eV)**	Electrons cannot overcome large bandgap under normal thermal energies; zero electrical conductivity

Table 3.4 — Band theory classification of electrical behavior.

Radius Ratio Rules — Limiting Geometry & Coordination Polyhedra

In ionic crystals, the stable geometric arrangement maximizes cation-anion attraction while avoiding anion-anion contact. The coordination number (CN) is governed by the **limiting radius ratio** $r_+ / r_- = r_c / r_a$.

1. Limiting Radius Ratio Rules Table

Radius Ratio (r_+ / r_-)	Coordination Number	Coordination Geometry	Mineral Lattice Prototype
	2	Linear	Gaseous molecules (BeF_2)
0.155 - 0.225	3	Planar Triangular	Boron trioxide (B_2O_3), Borates $[BO_3]^{3-}$
0.225 - 0.414	4	**Tetrahedral**	**Zinc Blende (ZnS)**, Wurtzite, BeO, CuCl
0.414 - 0.732	6	**Octahedral**	**Rock Salt (NaCl)**, MgO, FeO, KBr
0.732 - 1.000	8	**Body-Centered Cubic (Cubic)**	**Caesium Chloride (CsCl)**, CsBr, TlCl
1.000 (Equal sizes)	12	Close-Packed (FCC / HCP)	Pure Metals (Cu, Ag, Au)

Table 3.5 — Limiting radius ratios and coordination geometries.

2. Geometric Derivations of Limiting Values



Fig. 3.2 — Geometric derivation of limiting radius ratios: Octahedral square diagonal (0.414) and Cubic body diagonal (0.732).

- **Tetrahedral Limit ($r_+ / r_- = 0.225$):** Central cation touches 4 anions at tetrahedron corners: $\cos(109.5°/2) = \sin(54°44') = r_a/(r_c + r_a) \Rightarrow r_c / r_a = 0.225$.
- **Octahedral Limit ($r_+ / r_- = 0.414$):** Diagonal of equatorial square: $2(r_c + r_a) = \sqrt{2}\,(2\,r_a) \Rightarrow r_c / r_a = \sqrt{2} - 1 = \{0.414\}$.
- **Cubic Limit ($r_+ / r_- = 0.732$):** Body diagonal of cube: $2(r_c + r_a) = \sqrt{3}\,(2\,r_a) \Rightarrow r_c / r_a = \sqrt{3} - 1 = \{0.732\}$.

Classic Ionic Crystal Structures — Unit Cells, Packing & Void Occupancy

Ionic solids pack anions into close-packed lattices (FCC or HCP) with cations occupying tetrahedral or octahedral voids.

1. Master Structural Matrix of Ionic Solids

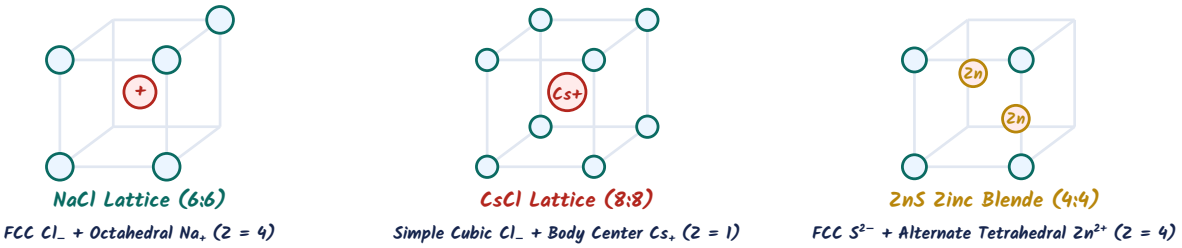


Fig. 3.3 — Classic ionic crystal lattices: NaCl (6:6), CsCl (8:8), and ZnS (4:4).

Structure Type	Anion Packing Array	Cation Position & Void Occupancy	Coord. No. (Cation : Anion)	Formula Units per Cell (Z)
Rock Salt (NaCl)	FCC array of Cl^- ions	Na^+ occupies **ALL Octahedral Voids**	**6 : 6**	**Z = 4** NaCl units
Caesium Chloride (CsCl)	Simple Cubic array of Cl^-	Cs^+ occupies central **Cubic Void**	**8 : 8**	**Z = 1** CsCl unit
Zinc Blende (Sphalerite ZnS)	FCC array of S^{2-} ions	Zn^{2+} occupies **ALTERNATE (50%) Tetrahedral Voids**	**4 : 4**	**Z = 4** ZnS units
Fluorite (CaF2)	FCC array of Ca^{2+} ions	F^- occupies **ALL (100%) Tetrahedral Voids**	**8 : 4** (Ca^{2+} is 8, F^- is 4)	**Z = 4** CaF2 units
Antifluorite (Na_2O)	FCC array of O^{2-} ions	Na^+ occupies **ALL (100%) Tetrahedral Voids**	**4 : 8** (Na^+ is 4, O^{2-} is 8)	**Z = 4** Na_2O units

Table 3.6 — Architectural parameters of standard ionic crystal lattices.

2. Transition Induced by Temperature and Pressure

KEY RESULT

Coordination Number Modulation: · **High Pressure:** Compresses lattice, increasing Coordination Number: **NaCl (6:6) →{High Pressure} CsCl (8:8)** · **High Temperature:** Expands lattice, decreasing Coordination Number: **CsCl (8:8) →{High Temperature (760 K)} NaCl (6:6)**

Crystal Imperfections & Defects — Stoichiometric, Non-Stoichiometric & F-Centres

Real crystals deviate from perfect thermodynamic order above 0 K. Defects alter electrical conductivity, optical colour, and mechanical strength.

1. Stoichiometric Point Defects



Fig. 3.4 — Crystal point defects: Schottky defect (equal ion vacancies) vs Frenkel defect (interstitial cation).

Defect Type	Physical Mechanism	Effect on Crystal Density	Conditions & Classic Examples
Schottky Defect	**Equal number of cations and anions missing** from normal lattice sites, creating pairs of stoichiometric vacancies	**Density DECREASES significantly** (mass decreases while cell volume is constant)	Favored in highly ionic compounds with **high coordination number** and **nearly equal ionic sizes**: NaCl, KCl, CsCl, KBr, AgBr
Frenkel Defect	**Smaller ion (usually cation) leaves normal lattice site** and occupies an **interstitial void**	**Density REMAINS UNCHANGED** (mass and volume both conserved)	Favored in compounds with **low coordination number** and **large size difference** ($r_- \gg r_+$): ZnS, AgCl, AgBr, AgI

Table 3.7 — Schottky versus Frenkel defects comparison.

KEY RESULT
The AgBr Special Case (Universal JEE Question): **Silver Bromide (AgBr) exhibits BOTH Schottky and Frenkel defects!** Radius of Ag+ is intermediate; it can fit into interstitial voids (Frenkel) or form stoichiometric ion pairs (Schottky).

2. Non-Stoichiometric Defects & F-Centres (Colour in Crystals)

- **Metal Excess Defect by Anion Vacancies (F-Centres):** When NaCl is heated in sodium vapor, Na atoms deposit on surface, forming Na+; Cl^- ions diffuse to surface to form NaCl. The liberated electrons diffuse into the lattice and occupy the anion vacancies. · An electron trapped in an anion vacancy is called an **F-Centre (Farbe centre / colour centre)**. · Trapped electrons absorb visible light and undergo transitions: **NaCl becomes Yellow, KCl becomes Violet/Lilac, LiCl becomes Pink!**
- **Metal Deficiency Defect:** Cation vacancies where missing charge is compensated by adjacent cations in higher oxidation states (e.g. Non-stoichiometric FeO exists as $Fe_{0.95}O$ containing both Fe^{2+} and Fe^{3+}).

Master Bonding Decision Algorithm — Step-by-Step Pathway for Any Molecule

A consolidated 5-step diagnostic workflow to solve any bonding, geometry, or polarity question in JEE Advanced.

1. The 5-Step Structural Decision Tree

1. **Step 1: Compute Total Valence Electrons & Draw Lewis Formula:** Check if octet is complete, expanded (Period 3+), or hypovalent. Calculate formal charges to pick the best resonance structure.
2. **Step 2: Determine Steric Number & Hybridization:** SN = σ-bonds + lone pairs. Apply Bent's rule if sp^3d (lone pairs eq, electronegative atoms ax). Check Drago's rule (if central atom is Period 3+ with lp and attached to H, no hybridization!).
3. **Step 3: Evaluate Back-Bonding / Bridge Bonding:** Check if an electron-deficient central atom (B, Be, Al) can receive dative π-backbonding from attached donor (F, O, N) or must dimerise via 3c-2e banana bonds (B_2H_6, BeH2) or 3c-4e halide bridges (Al_2Cl_6).
4. **Step 4: Predict VSEPR Geometry & Net Dipole Moment:** Place electron pairs to minimize repulsions. Deduce vector sum {μ}: cancel symmetric vectors (CO2, BF3, CCl4, SF6, XeF4 → μ = 0); distorted systems retain permanent dipole.
5. **Step 5: Apply Molecular Orbital Theory for Diatomics / Radicals:** For diatomics (1 to 20 electrons), write MO configuration. Check s-p mixing (≤ $14e^-$ vs > $14e^-$). Calculate Bond Order and identify unpaired electrons (paramagnetism).

2. Master Diagnostic Summary Table

Molecule	Central Hybrid.	VSEPR Geometry	Dipole Moment (μ)	Key Chemical Feature
XeF2	sp^3d (3 lp eq)	**Linear**	μ = 0	3 equatorial lone pairs cancel; non-polar
XeF4	sp^3d^2 (2 lp ax)	**Square Planar**	μ = 0	2 axial lone pairs oppose; non-polar
SF4	sp^3d (1 lp eq)	**See-Saw**	μ ≠ 0 (Polar)	Equatorial lone pair distorts axial bonds
ClF3	sp^3d (2 lp eq)	**T-Shaped**	μ ≠ 0 (Polar)	Two equatorial lone pairs compress F-Cl-F to 87.5°
I_3^-	sp^3d (3 lp eq)	**Linear**	μ = 0	Linear symmetrical triiodide anion
$[ICl_4]^-$	sp^3d^2 (2 lp ax)	**Square Planar**	μ = 0	Octahedral parent geometry with 2 axial lone pairs
$N(SiH_3)_3$	sp^2 (planar)	**Planar Trigonal**	μ = 0	pπ-dπ back-bonding destroys basicity
$O(SiH_3)_2$	Bent (widened)	**Bent (angle 144°)**	μ ≠ 0	2pπ(O) → 3dπ(Si) back-bonding widens angle

Table 3.8 — Diagnostic matrix for classic examination molecules.